

Spatial Alignment and Labeling for Immersive Content Organization

ABSTRACT

With the advancement of virtual reality technologies, maintaining reliable spatial alignment between virtual content and physical environments remains a critical challenge, particularly in large-scale interactive scenes. This work investigates the stability of spatial coordinates in MR and its application for immersive content organization. Through controlled experiments, we analyze factors affecting spatial anchor drift, including user movement speed, travel distance, and environmental complexity. Our findings reveal that the offset increases significantly with movement range, especially when navigating stairs or large open areas. Based on these obtained data, we propose two improvement methods: dense anchor placement and periodic anchor reloading using cloud-shared spatial data. Experimental results demonstrate that both approaches substantially reduce positional and rotational drift. Building on this improved alignment system, we introduce spatial labeling techniques including Areas of Interest (AOI) and Surface of Interest (SOI) for content organization, along with gesture-based interaction designs. Finally, we explore additional applications such as UI snapping to spatial anchors.

Index Terms: Spatial Alignment, Spatial Anchors, Spatial Labeling

1 INTRODUCTION & BACKGROUND

Virtual reality technology has evolved significantly since its inception in the 1960s, with applications expanding from flight simulators to education, medicine, design, and entertainment. While traditional VR experiences focus on visual and auditory feedback, incorporating haptic feedback through the combination of real objects and virtual scenes can greatly enhance immersive experience. However, maintaining consistency between real and virtual spaces requires stable spatial coordinates.

When using MR equipment, spatial coordinates can drift due to various internal and external factors. This loss of precision not only reduces realism but may also cause visual discrepancies between virtual models and real objects, leading to users' discomfort, fatigue, or even safety hazards from collisions with physical obstacles. Therefore, achieving reliable spatial alignment and exploring its applications is a topic worth investigating.

There are several existing alignment methods for large-scale MR scene. Due to time and equipment constraints, this project focuses exclusively on spatial anchor-based approaches.

2 EXPERIMENT

2.1 Experiment Design

This project first investigated whether spatial anchors drift during use, measuring positional and angular offset and identifying influencing factors. Based on Oculus spatial anchor operation principles, we selected the following variables for testing: walking speed, range of movement, travel route, headset stability, and system operation continuity. Using the headset's pass-through function, users could see real-world objects in areas not occluded by the Unity

modeling scene. With a spatial anchor placed at a certain position, recording the reference coordinates, Real-world object could serve as reference. The experimenter selected variable combinations through the UI panel, followed a fixed route, and then returned to the origin to place a new anchor at the same location. The system would calculate the spatial distance and the angular difference between the two anchors and recorded the data.

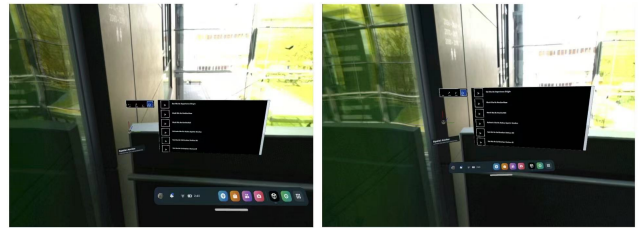


Figure 1: Reference anchor and how it drifted after user followed a fixed route

2.2 Results and Analysis

Each variable combination was repeated three times per data set and eight different combinations were tested. Results showed that when users moved only within a small area, anchor offset remained minimal even with fast movement and looking around. Manual placement differences contributed more to drift than system errors.

As movement expanded to straight-line corridor navigation, both positional and angular errors increased, particularly with faster travel and constant head rotation. When movement included stairs, errors increased significantly even at slow speeds due to safety considerations. The most dramatic increase occurred when moving around a large rectangular field, with position offsets reaching over 20 times as previous results. At this point, spatial anchors completely deviated from their original positions, causing the entire user interface to shift unpredictably, potentially overlapping with real walls or moving beyond accessible areas.

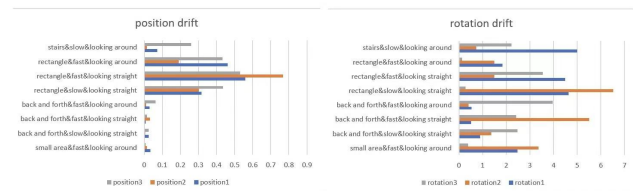


Figure 2: Experiment Results

When headset display was interrupted or internet connection lost during operation, spatial anchors would disappear upon re-entry since anchors were not uploaded or shared. For subsequent experiments testing improvements, we selected the combination most representative of real usage (rectangle route + walking fast + looking around), which also led to the largest errors.

3 SYSTEM IMPROVEMENT

Since Oculus provides Unity-accessible spatial anchor functions with local and cloud storage, and anchor positioning relies on point cloud technology, we designed two improvement approaches.

3.1 Improvement Design

3.1.1 Dense Anchors

Extra spatial anchors were placed approximately 5 centimeters below, above, in front of, and beside the original anchor position, forming a dense anchor group. The center anchor served as the reference during experiments.

3.1.2 Shared Anchors

After setting a spatial anchor at a fixed location, the experimenter saved anchor information locally or uploaded it to a server. Upon returning to the origin and observing offset, the anchor was reloaded once to correct its position before measuring the final offset.

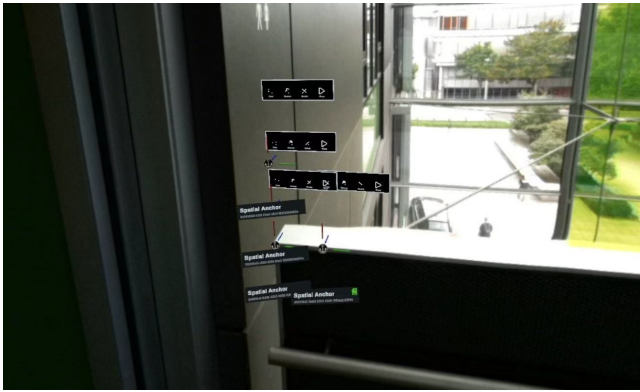


Figure 3: Dense Anchors

3.2 Results Analysis

3.2.1 Dense Anchors

Adding multiple densely arranged anchors significantly increased stability, drastically reducing both position and angular offsets.

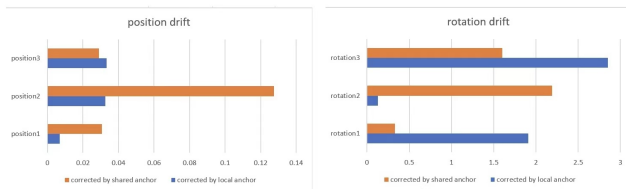


Figure 4: Experiment results while applying dense anchor

3.2.2 Shared Anchors

After correcting anchor positions using local or uploaded data, both position and angle offsets reduced significantly, closed to the small-offset levels observed in limited-range movement conditions. While point cloud technology inherently contains some uncertainty, corrected anchors demonstrated excellent stability, ensuring that system spatial stability remained unaffected by wide user movement.

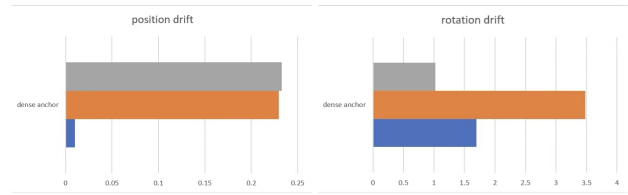


Figure 5: Experiment results while applying shared anchor

3.3 Improved System Design

Based on these findings, we designed the following spatial alignment system: Instead of the experimental 5cm placement, multiple spatial anchors may be placed on each edge or corner of objects to increase stability during object calibration. At large spatial scales, anchors are already relatively dense.

During system setup, a headset first places spatial anchors at desired locations and uploads them to a server. Subsequent users perform initial calibration after entering the scene, accessing the same wireless network to load anchors from the host. To minimize offset, each headset automatically reloads spatial anchors at fixed intervals (e.g., every 2 frames) without requiring user intervention, ensuring spatial stability without affecting user experience.

4 LABELING FOR CONTENT ORGANIZATION

To support large-scale immersive scenes, this project implemented spatial labeling, marking fixed areas where designers can place scene objects and users can drop objects they wish to display or use.

4.1 3D Labeling: Area of Interest (AOI)

AOI is a cubic volume defining three-dimensional space where objects can be placed. It serves to mark locations for objects or obstacles when scene modeling is incomplete, and to define object placement locations and sizes, helping controlling displayed model dimensions. After generating a new AOI via hand menu, users can grab it with hands to change the position or size.

The AOI UI panel allows users to select item type and material for that space, displayed in the AOI text box. The hand menu enables attaching AOI to the nearest spatial anchor or placing it independently.

4.1.1 Storage and Loading

Since devices only provide spatial anchor storage, AOI uses local storage. After placing AOIs and selecting save on hand menu, location, size, angle, type, and material information are stored locally as CSV files or uploaded to servers. Users can download and load AOI information from other devices. AOIs attached to spatial anchors use the anchor's UUID as location information; upon reloading, they locate that anchor and read its position. AOIs not attached to anchors use their own spatial coordinates.

4.1.2 Gesture Mode

When AOIs are large, grabbing and moving might be inconvenient. When the gesture control button on the hand menu is selected, users can use specific gestures to move the AOI plane directly in front of them to their current position.

4.2 2D Labeling: Surface of Interest (SOI)

SOI delineates a square flat area where users can place slides, images, screens for display, as well as flat objects, QR codes, and other functional 2D objects. SOI can also be stored, loaded, and uploaded as CSV files, utilizing spatial anchor locations.

For faster operation, when gesture mode is activated, users can generate SOI directly with specific gestures.

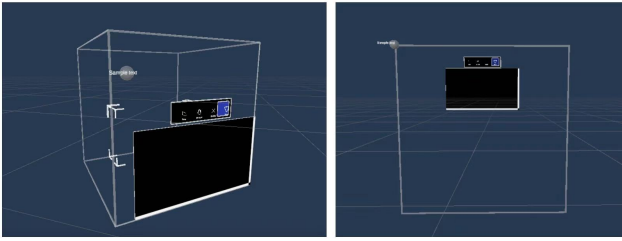


Figure 6: AOI and SOI prefab